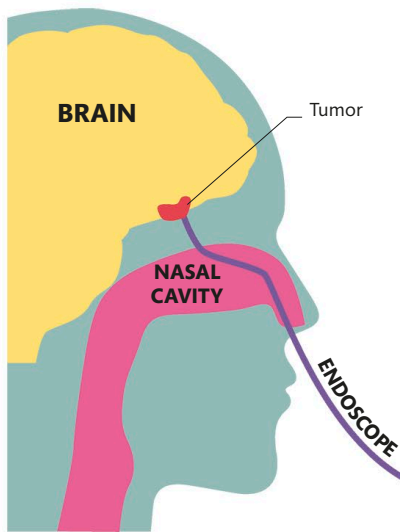




Transnasal brain surgery

Surgeons can now operate on some brain tumors through the nose. The procedure is much less invasive than a craniotomy, where the skull is opened and the brain exposed.

Brain tumors

A brain tumor is caused by cells that multiply in an abnormal way. It can occur in any part of the brain, from the intracranial space between the brain and the skull to deep within the brain itself. Tumors may be benign or malignant, and treatment varies accordingly.

There are approximately 130 types of brain tumors, and they are classified according to the kind of tumor or the area of the brain in which they grow. Some take years to develop, while others are much faster growing and more aggressive. Brain tumors can occur in people at any age or stage of life.

Locations and types

The most common types of brain tumors in adults are found in the cerebrum (see pp.28–29). About 24

percent start in the meninges—the membranes that surround and protect the brain and spinal cord. These tend to be easier to treat, if found early. Around 10 percent of brain tumors occur in the pituitary gland or pineal glands, which are surrounded by brain tissue.

In children, the picture is slightly different. Approximately 60 percent of childhood tumors occur in the cerebellum or brain stem. Only 40 percent arise in the cerebrum.

Dementia

Dementia is a term applied to a group of diseases associated with a decline in mental function that occurs most often in adults aged over 65. There are many different types of dementia.

Whether due to reduced blood flowing to the brain, a buildup of protein deposits, or other forms of damage, dementia in all its forms is a progressive disorder. Symptoms typically include mild forgetfulness, which may evolve into apathy or depression, reduced socialization, and loss of emotional control.

In later stages, a person with dementia may lose the ability to be compassionate or feel empathy, or to organize day-to-day activities. People with dementia often become very confused, not recognizing loved ones or knowing where they are. They might hallucinate, have

language difficulties, and need help with basic activities such as feeding or dressing themselves.

Diagnosis

While there is no cure for dementia, early diagnosis and treatment can slow the rate of mental decline. Brain scans highlight the areas of the brain most affected in an individual, and treatment can be tailored accordingly. The area most affected in Alzheimer's disease, for example, is the cortex. This part of the brain includes the hippocampus, where new memories are formed.

COMMON CAUSES OF DEMENTIA

Dementia can be caused by various disorders. Some of the most common are listed here.

Alzheimer's disease

A progressive condition in which bodies of proteins, called plaques, damage the brain.

Vascular dementia

Impaired blood flow to the brain, such as that caused by stroke, leads to a decline in function.

Lewy body dementia

Protein deposits in the brain's nerve cells affect thinking, memory, and motor control.

Frontotemporal dementia

A form that occurs in the front and sides of the brain, affecting behavior and language.

Parkinson's disease

Most Parkinson's sufferers develop dementia thought to be related to Lewy bodies.

Creutzfeldt-Jakob disease (CJD)

Rare, rapid, and fatal, this is caused by an infectious protein called a prion.